TTIC 31260: Algorithmic Game Theory

Final Project: Bluffing and Strategic Reticence in Prediction Markets

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1 Introduction

The global rise of prediction markets, particularly Polymarket and Kalshi, which have each amassed a weekly trading volume of more than 2.3 billion USD[1], has offered an alternative source of news compared to traditional news outlets. Main proponents argue that the mass aggregation of information from self-interested traders offer highly accurate predictions. However, is it of best interest for people to report their true beliefs? In other words, is the mechanism design incentive-compatible such that each trader reports their beliefs truthfully and immediately, i.e. *truthful betting*), or are there situations in which traders are instead incentivized to withhold information or even trade against their beliefs and 'bluff', in order to mislead other traders to receive better entry prices and profit more in the future?

As trust into prediction markets grow, this problem is arguably as important as ever, but Chen et al. already considered a simplified version of this problem in 2007 in their paper *Bluffing and Strategic Reticence in Prediction Markets* (2007) [1], which we will analyze in this report.

2 Summary

The authors initially claim that a double-sided auction, i.e. an auction where both buyers and sellers propose bids and asks is not incentive-compatible, as for example participants with full information will only leak their information gradually into the market, in order to achieve greater profit. To avoid this issue, the authors therefore consider market mechanisms employing so-called *automated market makers* (AMMs), which is an agent that is always willing to trade both sides and effectively subsidizes the market. Specifically, Hanson's market scoring rule market maker (MSR) [2], which subsidises participants by guaranteeing they cannot lose more than a fixed amount, and Pennock's dynamic parimutuel market (DPM) [3], in which participants compete for their share of the total money wagered by all participants.

Interestingly, AMMs have become fundamental in decentralized finance, most notably the foundation of the Uniswap protocol, which allows users to exchange tokens without the need of a traditional order book.

In the following, we consider a discrete random variable X with n mutually exclusive and exhaustive outcomes, which represents the prediction event.

[1] Odaily. "Polymarket's weekly notional trading volume surpasses Kalshi for the first time since last November." Bitget, 1 Mar. 2026, www.bitget.com/asia/news/detail/12560605229778.

2.1 Market Scoring Rule Market Maker (MSR)

Definition 2.1. Let $\mathbf{r} = (r_1, \ldots, r_n)$ denote a probability estimate over X. A *scoring rule* is a collection of functions $S = \{s_1(\mathbf{r}), \ldots, s_n(\mathbf{r})\}$, where $s_i(\mathbf{r})$ is the payment received if outcome i is realized. A scoring rule is called *proper*, if a risk-neutral agent maximizes their expected score by reporting their true beliefs.

Under this mechanism, any trader may update the probability estimate $\mathbf{r}$ from $\mathbf{r}^{\text{old}}$ to a new estimate $\mathbf{r}^{\text{new}}$, but pays $s_i(\mathbf{r}^{\text{old}})$ to the market maker in return for $s_i(\mathbf{r}^{\text{new}})$ from the market maker, where i is the outcome that is eventually realized.

Definition 2.2. The *logarithmic market scoring rule market maker* (LMSR) has scoring function $s_i(\mathbf{r}) := b \cdot \log(r_i)$ for some $b > 0$ and $i = 1, \ldots, n$.

Lemma 2.3. The logarithmic scoring rule is proper.

Proof. (This was not included in the paper) Let $\mathbf{p} = (p_1, \ldots, p_n)$ denote a trader's true beliefs and let $\mathbf{r} = (r_1, \ldots, r_n)$ denote their reported estimate. If we take the expectation under their true belief, we have

$$\mathbb{E}[s(\mathbf{p})] - \mathbb{E}[s(\mathbf{r})] = \sum_{i=1}^{n} p_i \cdot b_i \log(p_i) - \sum_{i=1}^{n} p_i \cdot b_i \log(r_i) = \sum_{i=1}^{n} p_i \cdot b_i \log\left(\frac{p_i}{r_i}\right) = \mathrm{D}_{\mathbf{KL}}(\mathbf{p} \| \mathbf{r}) \geq 0,$$

where D_{KL} is the Kullback-Leibner divergence, which is positive semi-definite. Therefore, a trader maximizes their expected scores by reporting their true beliefs. □

Setup. Let $\Omega = \{Y, N\}$ denote the state space representing the *Yes* and *No* outcome for a given event, where $\omega \in \Omega$ is chosen according to some prior $\mathbf{p}^0 = (p_Y^0, p_N^0)$, which is public to all participants. At each time t, players take turns in receiving a personal signal $c_i \in \mathbf{C}_i$ according to some conditional distribution $\mathbb{P}[c_i \mid \omega]$ for $\omega \in \{Y, N\}$ and updates their beliefs $\mathbf{r}_i^{t_1}$, which they can then decide to trade on.

The main theorem states that if the signals for players $i = 1, \ldots, N$ are attained conditionally independently on the outcome, then the LMSR mechanism is incentive-compatible.

Theorem 2.4 (Incentive Compatibility of LMSR)**.** *In a finite-player and finite-stage game with LMSR, if players receive conditionally independent signals, then a Perfect Bayesian Equilbrium of the game (each player's strategy is optimal given their beliefs, and beliefs are updated using Bayes' rule) is a strategy-belief pair, where each player reports their posterior probabilities in their first stage of play and all players believe that other players are truthful.*

This is exactly what we would want in a prediction market, as participants are upfront about their beliefs. However, this relies on the assumption of conditional independence and the authors show in the following example that in the case of dependent signals, we do not have this nice property anymore.

Example 2.5 (Bluffing under conditionally-dependent Signals)**.** Consider an Alice-Bob-Alice LMSR game where $\omega \in \{HH, HT \mid TH \mid TT\}$ represents whether both coins land heads. Alice and Bob each independently observe one coin flip, so their signals are conditionally dependent on the outcome, as given the full outcome, knowing one coin determines the outcome of the other coin. Let p denote the probability of heads for each coin.

If Bob trusts Alice's first-round move, Alice's best response would be to pretend to have heads regardless of her true signal: when Bob then observes heads on his own coin, he moves the probability

of HH to 1, which was shown by Lemma 1 in the paper that states that a player reports their true belief if it is the last opportunity to report. Then Alice corrects the price. Therefore, truthful betting cannot be an equilibrium. Instead in the equilibrium, Alice mixes between honest reporting and bluffing as if her coin is heads ($\hat{H}$) and truthfully betting with probability

$$1 + \frac{p}{\big(1-(1-p)^{-1/p}\big)(1-p)},$$

and otherwise playing $\hat{H}$.

2.2 Dynamic Parimutuel Market (DPM)

Definition 2.6. A *dynamic parimutuel market* (DPM) is a subsidized market mechanism over n securities each corresponding to an outcome of X. Let $\mathbf{q} = (q_1, \ldots, q_n)$ denote a vector of the oustanding shares. The cost function (total money wagered) and instantaneous price of security i are given by

$$C(\mathbf{q}) = \kappa\sqrt{\sum_{j=1}^{n} q_j^2}, \qquad p_i(\mathbf{q}) = \frac{\kappa q_i}{\sqrt{\sum_{j=1}^{n} q_j^2}},$$

for some parameter $\kappa > 0$. Unlike LMSR, where a participant provides a probability estimate, in DPM the market probability $\pi_i(\mathbf{q})$ is derived from the share vector. The implied market probability of outcome i is

$$\pi_i(\mathbf{q}) = \frac{q_i^2}{\sum_{j=1}^{n} q_j^2}.$$

Under this mechanism, a trader who moves the share vector from $\mathbf{q}^{\text{old}}$ to $\mathbf{q}^{\text{new}}$ pays the market maker $C(\mathbf{q}^{\text{new}}) - C(\mathbf{q}^{\text{old}})$. If outcome i is realized and the final share vector is $\mathbf{q}^f$, the payoff per winning share is the the total money wagered divided by the quantity of the winning share, i.e.

$$o_i = \frac{C(\mathbf{q}^f)}{q_i^f} = \frac{\kappa\sqrt{\sum_j (q_j^f)^2}}{q_i^f}. \tag{1}$$

The main result from the paper was that under certain conditions in this market, a player would withhold their information.

Example 2.7 (Strategic Withholding under a Perfectly Informed Oracle)**.** Consider a two-outcome $\{X, Y\}$ DPM with initial share vector $\mathbf{q}^0 = (1, 1)$, i.e. implied probability $\frac{1}{2}$ for outcome Y. Suppose Alice holds posterior $p > 1/2$ for outcome Y and that there exists a perfectly informed oracle of the true outcome that can trade after Alice.

Let x and g denote the quantities of Y-shares purchased by Alice and the Oracle respectively. Alice's expected utility (computed from the payoff (1)) is

$$\kappa\left(px\sqrt{\frac{(1+x+g)^2+1}{1+x+g}} - p\sqrt{(1+x)^2+1} - \sqrt{2}\right).$$

Assume outcome Y occurs, then we consider the limit $g \to \infty$ as the oracle will purchase as much as possible and Alice's expected utility simplifies to

$$\kappa\Big(px - p\sqrt{(1+x)^2+1} + \sqrt{2}\Big).$$

Maximizing over x using first-order condition yields

$$x^* = \max\left(0,\ \sqrt{\frac{p^2}{1-p^2}} - 1\right),$$

which is positive only when $p > 1/\sqrt{2} \approx 0.707$. Thus, the presence of a perfectly informed trader discourages Alice from revealing her beliefs, unless Alice holds a posterior greater than $\frac{1}{\sqrt{2}}$. When she does reveal her information, she moves the outstanding shares from $(1,1)$ to $(1, 1+x^*)$, which has implied probability of p^2 and is still less than her posterior p.

3 Extensions

In this section, we consider extensions of this paper. We ask ourselves two questions, namely how strict is the assumption of conditional independence in the LMSR market in Theorem 2.4, i.e. are there looser assumptions to ensure incentive-compatibility? Secondly, we want to consider a well-informed oracle, but not perfectly oracle as in Example 2.7 and determine how informed the oracle has to be such that Alice withholds information.

3.1 Conditions to ensure LMSR incentive-compatibility

To determine the strictness of the conditional independence assumption in Theorem 2.4, we consider the proof for the Alice-Bob-Alice game, where conditional independence is utilized to simplify the difference in Alice's expected utility between truthful betting and bluffing into a Kullback-Leibler divergence similar to Lemma 2.3 and since it is non-negative, it shows that Alice is better off betting truthfully.

Let c_A be Alice's signal, c_B be Bob's signal, and $\mathbf{p}^0$ be the initial market probability. If Alice bluffs according to some fake posterior $r^0 \neq \mathbb{P}(\omega|c_A)$ in Stage 1, Bob will update the market in Stage 2 to $\mathbb{P}(\omega|r^0, c_B)$. In Stage 3, Alice corrects the market to the true joint posterior $\mathbb{P}(\omega|c_A, c_B)$. We can compare her expected utility from truthful betting (EU_1) against her expected utility from bluffing (EU_2). The utility that Alice obtains from truthful betting is just moving the probability from $\mathbf{p}^0(\omega)$ to $\mathbb{P}(\omega \mid c_A)$, i.e. obtaining the payoff $\log(\mathbb{P}(\omega \mid c_A)) - \log(\mathbf{p}^0(\omega))$

$$EU_1 = \mathbb{E}_{c_B,\omega|c_A}\left[\log(\mathbb{P}(\omega \mid c_A)) - \log(\mathbf{p}^0(\omega))\right].$$

Similarly, when Alice bluffs, she first moves the probability from $\log(r^0(\omega))$ to $r^0(\omega)$ and in her third turn from $\mathbb{P}(\omega \mid r^0, c_B)$ to $\mathbb{P}(\omega \mid c_A, c_B)$.

$$EU_2 = \mathbb{E}_{c_B,\omega|c_A}\left[\log(r^0(\omega)) - \log(\mathbf{p}^0(\omega)) + \log(\mathbb{P}(\omega \mid c_A, c_B)) - \log(\mathbb{P}(\omega \mid r^0, c_B))\right],$$

By taking the difference $EU_1 - EU_2$

$$\begin{aligned} EU_1 - EU_2 &= \mathbb{E}_{c_B,\omega|c_A}\left[\log(\mathbb{P}(\omega|c_A)) - \log(r^0(\omega)) - \log(\mathbb{P}(\omega|c_A, c_B)) + \log(\mathbb{P}(\omega|r^0, c_B))\right] \\ &= \mathbb{E}_{c_B,\omega|c_A}\left[\log\left(\frac{\mathbb{P}(\omega|c_A)\mathbb{P}(\omega|r^0, c_B)}{\mathbb{P}(\omega|r^0)\mathbb{P}(\omega|c_A, c_B)}\right)\right]. \end{aligned}$$

If the signals are conditionally independent, then this can be further simplified into an expression with KL-divergence. However, at its core we only would need the expression $\mathbb{E}_{c_B,\omega|c_A}\left[\log\left(\frac{\mathbb{P}(\omega|c_A)\mathbb{P}(\omega|r^0,c_B)}{\mathbb{P}(\omega|r^0)\mathbb{P}(\omega|c_A,c_B)}\right)\right]$ to be non-negative, in order to achieve incentive-compatibility.

3.2 Withholding information in the presence of well-informed trader

Instead of a perfectly-informed oracle, we want to only consider a well-informed oracle, i.e. knows the true outcome with some probability p^{oracle}. Ideally, we want to find, some expression for Alice's conviction p^A at which Alice would (partially) reveal her information and enter a trade, i.e. $x^* > 0$. Alternatively, one could also consider the threshold p^{oracle} at which Alice reveals all of her information and moves the implied probability to p^A.

We use the same setup as Example 2.7. Assume the initial share vector is $\mathbf{q}^0 = (1, 1)$. Alice purchases x shares of Y and the Oracle trades after Alice, purchasing g shares of Y to move the market probability to their belief q. By the DPM probability function, the market probability of Y after the Oracle's trade is:

$$p^{\text{oracle}} = \frac{(1+x+g)^2}{(1+x+g)^2+1}$$

Rearranign for the total shares of Y outstanding yields $1 + x + g = \sqrt{\frac{p^{\text{oracle}}}{1-p^{\text{oracle}}}}$. If outcome Y is realized, the payoff for each winning share o_Y is the total money wagered divided by the quantity of winning shares. Substituting this gives

$$o_Y = \kappa \frac{\sqrt{(1+x+g)^2+1}}{1+x+g} = \kappa \frac{\sqrt{\frac{p^{\text{oracle}}}{1-p^{\text{oracle}}}+1}}{\sqrt{\frac{p^{\text{oracle}}}{1-p^{\text{oracle}}}}} = \frac{\kappa}{\sqrt{p^{\text{oracle}}}}$$

We want to maximize Alice's expected utility, which is her expected payout minus her initial cost to purchase x

$$EU_A(x) = p^A \left(x \frac{\kappa}{\sqrt{p^{\text{oracle}}}} \right) - \kappa \left(\sqrt{(1+x)^2+1} - \sqrt{2} \right)$$

We determine Alice's optimal share purchase quantity x^* using first-order condition $\frac{\partial EU_A}{\partial x} = 0$

$$\frac{\kappa p^A}{\sqrt{p^{\text{oracle}}}} - \kappa \frac{1+x}{\sqrt{(1+x)^2+1}} = 0 \implies \frac{1+x}{\sqrt{(1+x)^2+1}} = \frac{p^A}{\sqrt{p^{\text{oracle}}}}$$

Rearranging for x gives

$$x^* = \max\left(0, \frac{p^A}{\sqrt{p^{\text{oracle}} - (p^A)^2}} - 1\right)$$

and Alice will only participate in the market and reveal information if her optimal purchase quantity is strictly positive, i.e. $p^A > \sqrt{\frac{p^{\text{oracle}}}{2}}$, which coincides for the case $p^{oracle} = 1$ in Example 2.7.

4 Conclusion

Though this paper did not cover content that we explicitly cover in class, concepts like incentive-compatibility still translated in this context. I also read through Chapter 26.4 in the textbook.

We considered some extensions from the original paper and found interesting conditions in a DPM market, when to reveal information trading when trading with a better informed trader. Also interestingly I find, is that recently Polymarket introduced a feature to 'sponsor' markets which allows anybody to effectively subsidize market makers, which may lead to more incentive-compatible behavior.

References

[1] Yiling Chen, Daniel M. Reeves, David M. Pennock, Robin D. Hanson, Lance Fortnow, and Rica Gonen. Bluffing and strategic reticence in prediction markets. `https://bpb-us-e1.wpmucdn.com/sites.harvard.edu/dist/b/845/files/2025/01/bluffing-long-appendix.pdf`, 2007.

[2] Robin Hanson. Logarithmic market scoring rules for modular combinatorial information aggregation. Technical report, Department of Economics, George Mason University, January 2002.

[3] David M. Pennock. A dynamic pari-mutuel market for hedging, wagering, and information aggregation. In *Proceedings of the 5th ACM Conference on Electronic Commerce (EC '04)*, New York, New York, USA, May 2004. Association for Computing Machinery.